SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA1402 **COURSE NAME: Compiler Design**

COURSE OUTCOME COVERED

CO4: Examine intermediate code generation techniques to enhance code optimization (BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:50**

Annexure A

CLASS TEST

Code of Conduct:

I Akshaya A(192421222) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Akshaya A

Q. No	Understanding of Concepts (25)	Explanation (25)	Application (20)	Organization(15)	Accuracy(10)	Total Score (out of 100)
1	/4	/4	/4	/4	/4	
2	/4	/4	/4	/4	/4	
3	/4	/4	/4	/4	/4	
4	/4	/4	/4	/4	/4	
5	/4	/4	/4	/4	/4	
OVERALL RUBRICS VALUE						
	/4	/4	/4	/4	/4	
	/25	/30	/20	/15	/10	/100

Annexure A

CLASS TEST

Q1. Construct three address code for the following

if((a < b) and ((c < d) or (a > d))) then

z=x + y *z

else

z = z + 1

Q2.Translate (a < b) and (c < d) or (e < f) into three address statements using back patching.

Q3. Construct three address code using the following production for Boolean expression (a > b) or (c < d) or (e < f)

B → B1 || B2

B → B1 && B2

B → !B1

B → E1 relop E2

B → true

B → false

Q4. Generate the three address code for the following program fragment-

while (A < C and B > D) do

if A = 1 then C = C + 1

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COURSE: CSA1402 COMPILER DESIGN

CO4_AT_1_CLASS TEST

1. Construct Three Address code,

if ((a<b) and ((c<d) or (a>d))) then

z = x + y * z

else

z = z + 1

38/40

Ans: Given: if ((a<b) and ((c<d) or (a>d))) then

z = x + y * z

else

z = z + 1

Conditions: a<b

c<d

a>d

Labels: L1, L2, L3, L4, L5

L1:

if a<b goto L2

goto L5

L2:

if c<d goto L4

goto L3

L3 :

if $(a>d)$ goto L4

goto L5

Three Address code

L4 $\Rightarrow$ $z = x + y * z$

$\therefore$ L4:

$t1 = y * z$

$t2 = x + t1$

$z = t2$

goto L6

L5 $\Rightarrow$ $z = z + 1$

$\therefore$ L5 :

$t3 = z + 1$

$z = t3$

2) Translate $(a<b)$ and $(c<d)$ or $(e<f)$ into three address statements using Back patching.

Ans:

$[(a<b)$ AND $(c<d)]$ OR $(e<f)$

if $(a<b)$ goto 3

goto 5

if $(c<d)$ goto 7

goto 5

if $(e<f)$ goto 7

goto 8

7. True

8. False

3. Construct three address code using the following production for boolean expression $(a>b)$ or $(c<d)$ or $(e<f)$.

$B \rightarrow B_1 \mid\mid B_2$

$B \rightarrow B_1 \ \&\& \ B_2$

$B \rightarrow \ ! \ B_1$

$B \rightarrow E_1 \ \text{rel.op} \ E_2$

$B \rightarrow$ true

$B \rightarrow$ false

Ans. L1: if $(a>b)$ goto LTRUE

goto L2

L2:

if $(c<d)$ goto LTRUE

goto L3

L3: if (e < f) goto LTRUE

goto LFalse

4) Generate the three address code for the following program fragment - while (A < C and B > D) do

if A = 1 then C = C + 1

else while A <= D do

A = A + B

Ans:

```
L1:
    if A < C goto L2
    goto LEND

L2:
    if B > D goto L3
    goto LEND

L3:
    if A == 1 goto L4
    goto L5

L4:
    t1 = C + 1
    C = t1
    goto L1

L5:
    if A <= D goto L6
    goto L1
L6: t2 = A + B
    A = t2
    goto L5
LEND:
```

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	15%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps